

The Margin

Eight number patterns that decide Domain 3 stems, three fresh variants each, answers sealed at the back. Twenty minutes. The goal is that the multiplication happens in the margin in ten seconds, so the 120-second stem is spent on the tell, not the sum.

1. **Belt tax · objective 3.1**
2. **Blended cost against a ceiling · objective 3.3**
3. **p95 plus an addition, against an SLA · objective 3.3 / 3.4**
4. **Skip-RAG threshold and pages · objective 3.5 / 3.8**
5. **Cache-read equivalents and minimums · objective 3.8**
6. **Batch feasibility: window first, then time · objective 3.8 (D1 crossover)**
7. **Overlap ratio · objective 3.5**
8. **Cardinality: series per label set · objective 3.4**

THE MARGIN · HOW TO DRILL

Eight patterns, three variants each

Each pattern gives the method in one line and one worked example, then three variants with a blank margin. Write the expression, then the number, then the **verdict word** (fits / breaches / trim / cache / retrieve). The verdict is what the option letter tests; the number is how you get there without guessing. Answers are sealed at the back.

◎ THE HABIT

Every number-decided stem hides one of these eight shapes. Recognise the shape first, then compute; a stem that gives you three numbers wants you to combine exactly two of them.

1 · Belt tax

Objective 3.1 · **Method:** idle schemas × tokens per schema × calls per day — then say the verdict: trim, re-add when a log line asks.

✎ WORKED

6 idle × 250 × 40,000 = 60,000,000 (60M) tokens/day.

#	Variant	Expression · number · verdict
1.1	Seven tools show 0 calls last week; schemas average 240 tokens; 25,000 calls a day. Daily belt tax?	
1.2	count_tokens: 3,100 with 11 tools, 1,300 with the 4 that carry traffic; 60,000 calls/day. Daily tax of the idle 7?	
1.3	Two lookalike tools at 270 tokens each are merged into one 300-token tool; 40,000 calls/day. Tokens saved per day?	

2 · Blended cost against a ceiling

Objective 3.3 · **Method:** Σ (share of volume × cost per case) — compare with the ceiling BEFORE touching any dial.

✎ WORKED

$0.02 \times ₹38 + 0.98 \times ₹1.20 = ₹1.94 \rightarrow$ under a ₹6 ceiling; no downgrade needed.

#	Variant	Expression · number · verdict
2.1	Agent tier ₹42/case at 3% of volume; drafts ₹1.10 at 97%. Ceiling ₹6. Blended, and verdict?	
2.2	Refunds ₹9.80 at 9%; drafts ₹0.90 at 55%; lookups ₹0.20 at 36%. Ceiling ₹6.	

#	Variant	Expression · number · verdict
2.3	A proposal moves ALL routes to the large model at ₹9.40/case. Ceiling ₹6. Verdict?	

3 · p95 plus an addition, against an SLA

Objective 3.3 / 3.4 · **Method:** current p95 + the added latency, compared with the SLA — at p95, never at the mean.

WORKED

$1.9\text{ s} + 0.2\text{ s} = 2.1\text{ s} > 2.0\text{ s SLA} \rightarrow$ reject on this path.

#	Variant	Expression · number · verdict
3.1	Router p95 1.75 s; SLA 2 s; proposal adds a 300 ms reranker.	
3.2	SOP path p95 2.8 s; SLA 4 s; proposal adds 200 ms rerank + 350 ms for k=40.	
3.3	Mean 310 ms, p95 2.9 s, SLA 3 s end-to-end; draft p95 adds 1.2 s.	

4 · Skip-RAG threshold and pages

Objective 3.5 / 3.8 · **Method:** corpus tokens vs ~200k; ~400 tokens per page; stable + always-used decides with it.

WORKED

38k tokens, quarterly changes, every call $\rightarrow$ under 200k $\rightarrow$ no RAG, cache it.

#	Variant	Expression · number · verdict
4.1	Rate card + templates: 95 pages, quarterly, on nearly every call. Tokens, and RAG or not?	
4.2	400 SOPs averaging 3,500 tokens. Tokens, and RAG or not?	
4.3	A 150k-token pack, stable, but read by 30% of calls. Cache or retrieve?	

5 · Cache-read equivalents and minimums

Objective 3.8 · **Method:** cached tokens $\times \sim 0.1$ — and check the prefix clears the tier minimum (1,024–4,096; Haiku 4.5 = 4,096).

WORKED

$45,000 \times 0.1 = 4,500$ token-equivalents; no extra turn.

#	Variant	Expression · number · verdict
5.1	An 8,000-token preamble cached with a 92% hit rate. Average billed equivalents per call?	
5.2	A 60k-token pack cached vs retrieving top-20 (3,600 tokens + 180 ms). Compare.	
5.3	A 2,000-token preamble marked for caching on Haiku 4.5. Hits?	

6 · Batch feasibility: window first, then time

Objective 3.8 (D1 crossover) · **Method:** does the corpus fit ANY window? then per-case latency × cases vs the batch window.

WORKED

1.4M tokens fits no window → retrieval; $180 \text{ ms} \times 400 = 72 \text{ s} \ll 4 \text{ h}$.

#	Variant	Expression · number · verdict
6.1	Nightly batch, 400 cases, retrieval 180 ms/case, window 4 hours.	
6.2	1,200 cases; per-case hybrid + rerank 650 ms; window 30 minutes.	
6.3	A 45k rate card in every case's prompt for 400 cases, cached. Token-equivalents for the card across the batch?	

7 · Overlap ratio

Objective 3.5 · **Method:** $\text{overlap} \div \text{chunk size} = \text{duplicated fraction}$; index growth $\approx 1 + \text{ratio}$.

WORKED

$40 \div 500 = 8\%$ duplicated; index $\approx 1.08\times$.

#	Variant	Expression · number · verdict
7.1	Chunks of 500 tokens with 400 overlap.	
7.2	Chunks of 350 tokens with 40 overlap.	

#	Variant	Expression · number · verdict
7.3	3,600 chunks with 25% overlap; tokens indexed if the average chunk is 300?	

8 · Cardinality: series per label set

Objective 3.4 · **Method:** multiply the distinct values of every label; ids make it explode.

WORKED

labels {stage×4, carrier×3, route×4} = 48 series; {consignment_id} = 40,000 new series a day.

#	Variant	Expression · number · verdict
8.1	Labels: stage (4) × carrier (3) × route (4) × model tier (3).	
8.2	Labels: stage (4) × consignment id (40,000/day).	
8.3	A 10% sample of consignment ids as a label, 4 stages.	

SEALED

THE ANSWERS

Twenty-four variants. Do not turn until the margin is full.

The answers — expression, number, verdict

#	Pattern	Expression	Number · verdict
1.1	Belt tax	7 idle tools × 240 tokens × 25,000 calls/day	42,000,000 tokens/day (42.0M)
1.2	Belt tax	(3,100 – 1,300) × 60,000	108,000,000 tokens/day (108M)
1.3	Belt tax	(540 – 300) × 40,000	9,600,000 tokens/day (9.6M) — and the confusion bill drops too
2.1	Blended cost against a ceiling	0.03×₹42 + 0.97×₹1.1	₹2.33 blended
2.2	Blended cost against a ceiling	0.09×₹9.8 + 0.55×₹0.9 + 0.36×₹0.2	₹1.45 blended
2.3	Blended cost against a ceiling	1.00 × ₹9.40	₹9.40 — breaches the ceiling by ₹3.40; tier by consequence instead
3.1	p95 plus an addition, against an SLA	1.75 + 0.30	2.05 s > 2.0 s — breach; reject on this path
3.2	p95 plus an addition, against an SLA	2.8 + 0.2 + 0.35	3.35 s < 4 s — fits, but k=40 buys no measured accuracy: take the rerank, refuse the k
3.3	p95 plus an addition, against an SLA	2.9 + 1.2	4.1 s > 3 s at p95 — the mean was never the number; fix the retrieve tail
4.1	Skip-RAG threshold and pages	95 × ~400	≈ 38,000 tokens → under 200k, stable, always used → no RAG; cache
4.2	Skip-RAG threshold and pages	400 × 3,500	1,400,000 tokens → far over 200k → RAG (chunk, hybrid, rerank)
4.3	Skip-RAG threshold and pages	150k < 200k, but 30% ≠ always	retrieve — 70% of calls would carry 150k cached tokens for nothing
5.1	Cache-read equivalents and minimums	0.92 × 800 + 0.08 × 8,000	≈ 1,376 token-equivalents
5.2	Cache-read equivalents and minimums	60,000 × 0.1 vs 3,600 + a hop	6,000 vs 3,600 — cache wins on turns and under-fetch risk; the token column alone misleads
5.3	Cache-read equivalents and minimums	2,000 < 4,096 minimum	zero — never cached; enlarge the prefix or stop expecting hits
6.1	Batch feasibility: window first, then time	0.18 × 400	72 s against 14,400 s — trivially fits
6.2	Batch feasibility: window first, then time	0.65 × 1,200	780 s = 13 min against 1,800 s — fits with headroom
6.3	Batch feasibility: window first, then time	400 × 4,500	1,800,000 equivalents — vs 400 × 45,000 = 18M uncached
7.1	Overlap ratio	400 ÷ 500	80% duplicated; index ≈ 1.8× — the dial is turned too far

#	Pattern	Expression	Number · verdict
7.2	Overlap ratio	$40 \div 350$	$\approx 11\%$ duplicated; index $\approx 1.11\times$ — seam insurance, priced
7.3	Overlap ratio	$3,600 \times 300 \times 1.25$	1,350,000 tokens indexed
8.1	Cardinality: series per label set	$4 \times 3 \times 4 \times 3$	144 series — fine
8.2	Cardinality: series per label set	$4 \times 40,000$	160,000 series a day — the trap
8.3	Cardinality: series per label set	$4 \times 4,000$	16,000 series a day — still the trap, slower

THE EIGHT SHAPES IN ONE BREATH

1. Belt tax = idle $\times$ per-schema $\times$ daily calls. Blended cost = Σ share $\times$ price. Both are linear; sanity-check the order of magnitude.
2. Latency adds at p95, and the SLA is compared at p95 — the mean is never in the sum.
3. ≈ 400 tokens per page; $\sim 200k$ tokens ≈ 500 pages is the skip-RAG line — with 'stable' and 'always-used' as the other two conditions.
4. Cache reads $\approx 0.1\times$; minimum cacheable prefix 1,024–4,096 by tier (Haiku 4.5 = 4,096).
5. Feasibility before taste: does it fit the window? Then time $\times$ cases vs the batch window.
6. Overlap $\div$ chunk = duplicated fraction. Labels multiply; ids explode.